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Software Requirements Specification (SRS)

I. Introduction

Chess has been one of the most popular strategy games for centuries, and with the advancement of artificial intelligence (AI), it has become even more engaging and educational. This document outlines the requirements for an AI-integrated online chess platform, designed to provide players with a seamless and enriching chess experience. The platform will support both player-versus-player (PvP) and player-versus-environment (PvE) modes, where users can challenge AI opponents of varying skill levels. Additionally, it will include an AI-powered training mode that provides real-time suggestions and post-game analysis to help players improve their strategies.

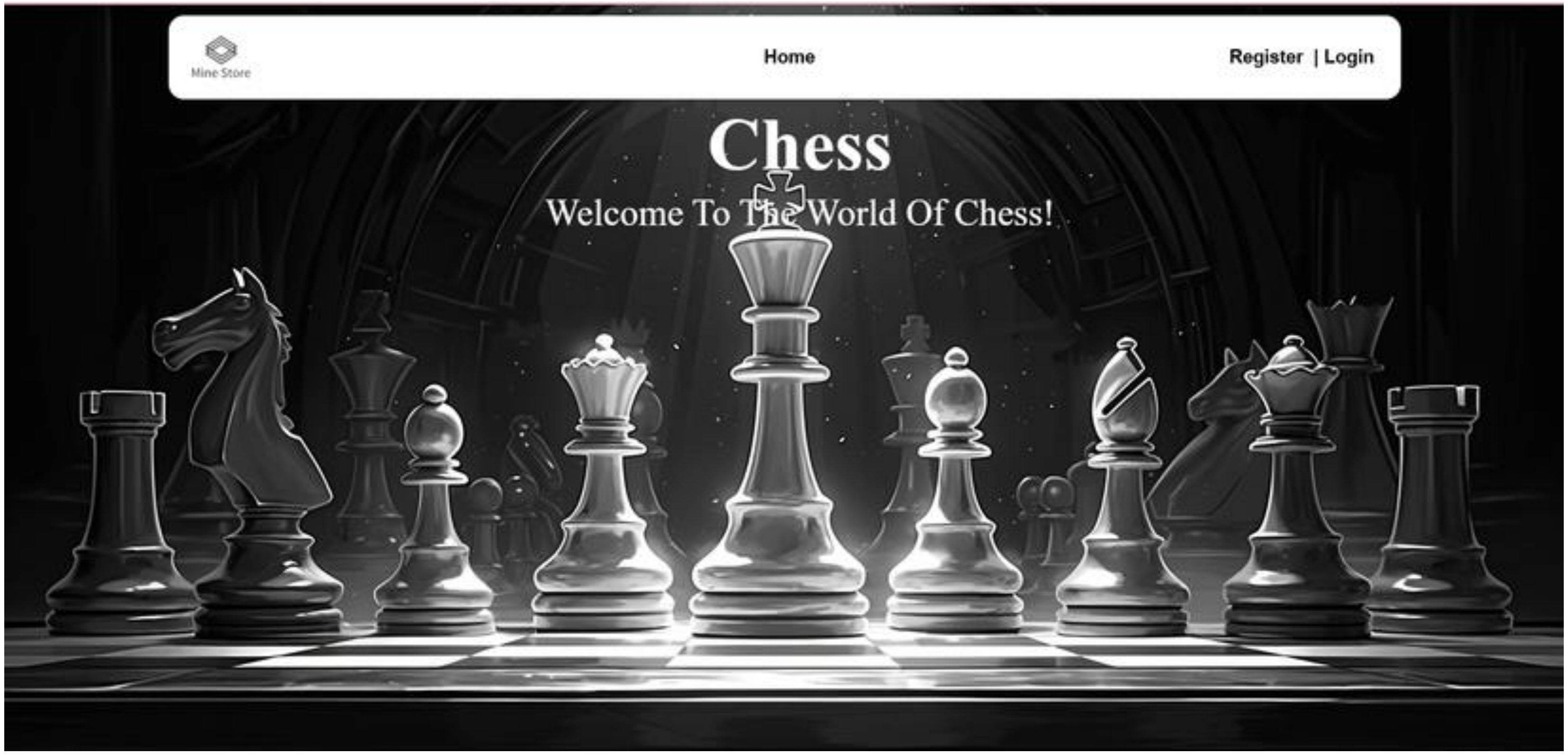
The primary audience for this platform includes chess enthusiasts of all levels, from beginners who need guided assistance to advanced players looking for challenging AI matches and in-depth game analysis. Developers and system administrators will also find this document useful in understanding the technical specifications required to build and maintain the platform.

The platform will feature user registration, matchmaking, ranking systems, and an intuitive interface optimized for desktop and mobile devices. AI integration will be based on advanced chess engines like Stockfish, ensuring high-quality gameplay and analysis. Security measures, such as encrypted data storage and anti-cheating mechanisms, will be implemented to create a fair and competitive environment.

This document provides a structured approach to defining the functional and non-functional requirements of the system, ensuring a clear vision for its development and deployment.

II. Game images

Main interface of our chess website

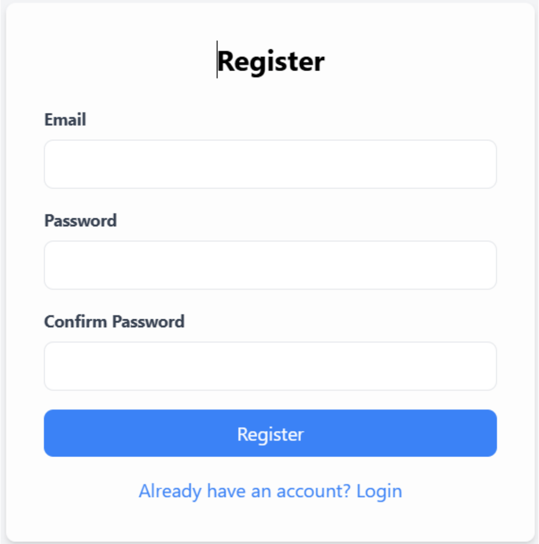


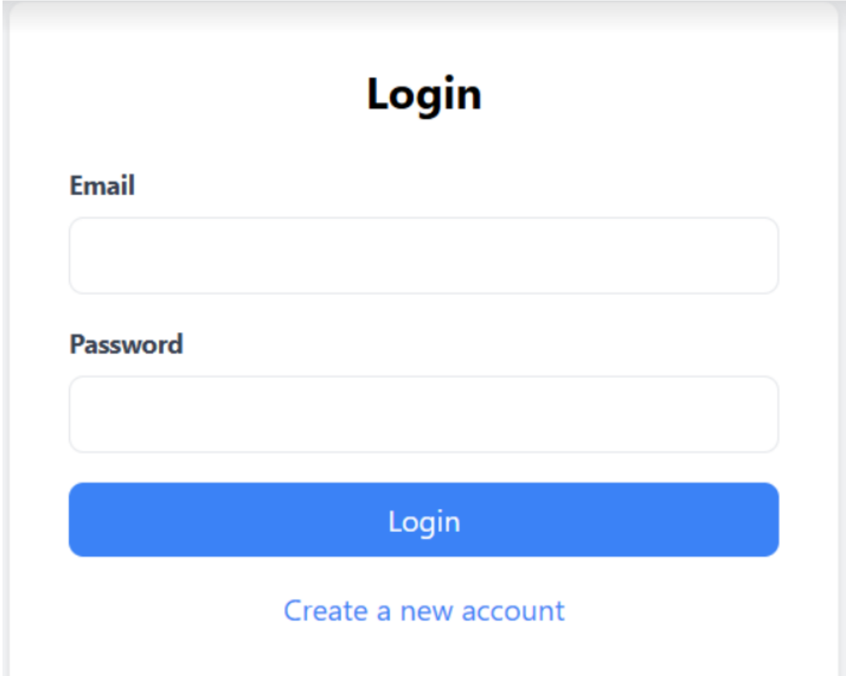
On our site you can choose to play in many forms.

Register/Login

Players can register an account using email and password to create an account.

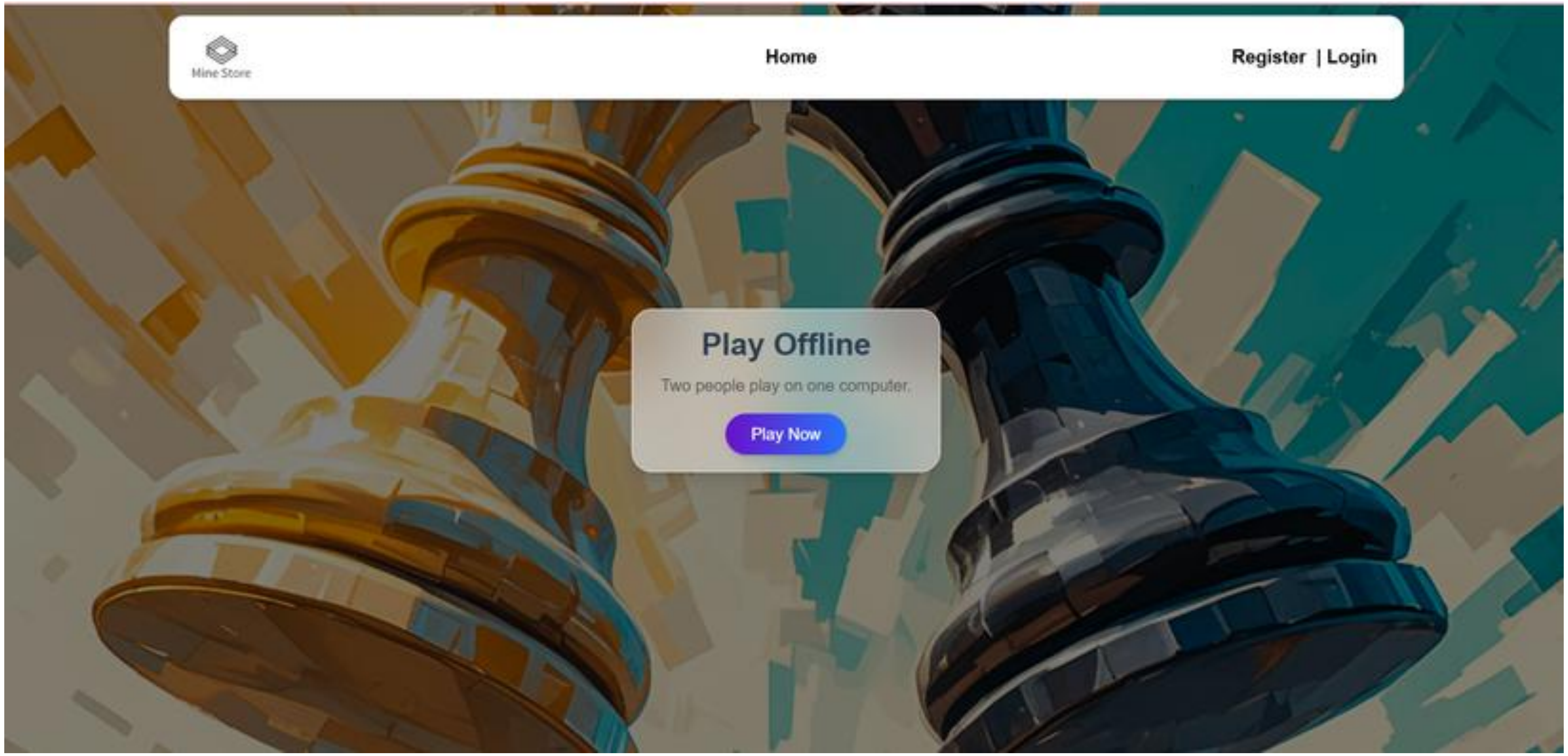
The system authenticates accounts and manages login sessions.





A.Play Offline

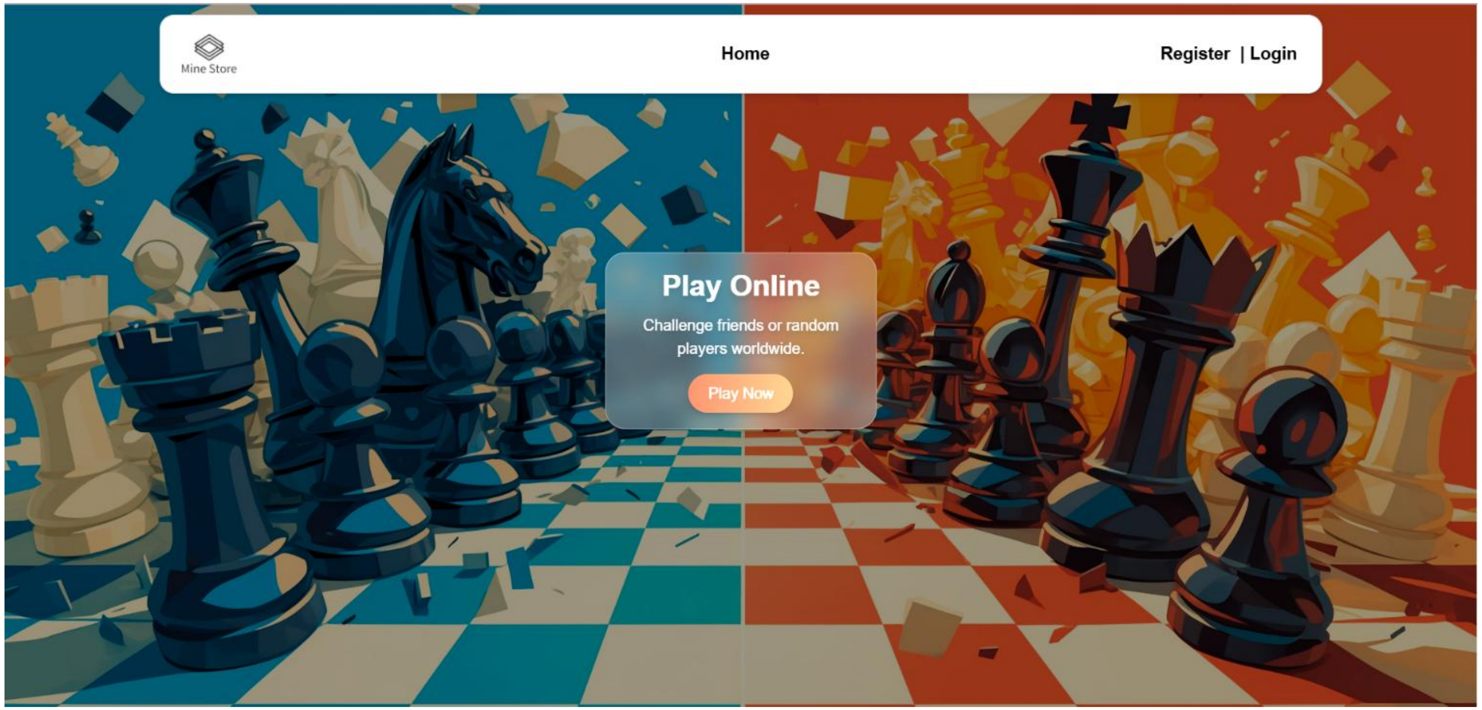
Players can play with friends on the same machine.



B.Play Online with Other Players

Players can randomly match up or invite friends to the room.

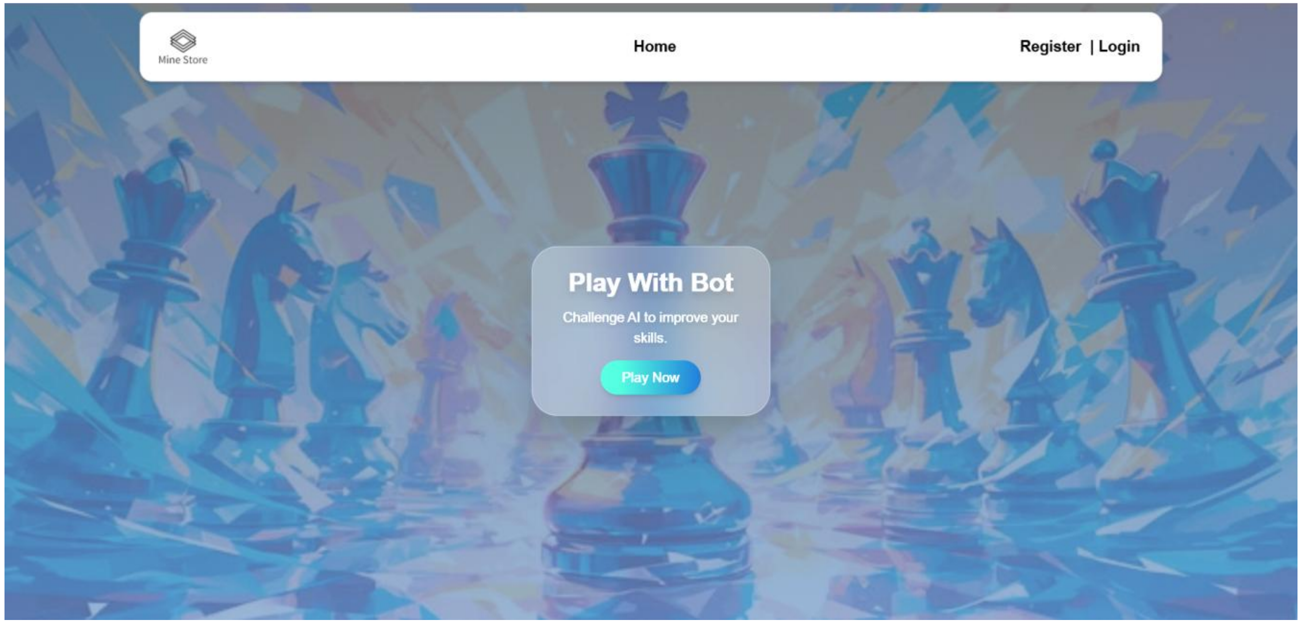
Use WebSocket to transmit moves in real time.



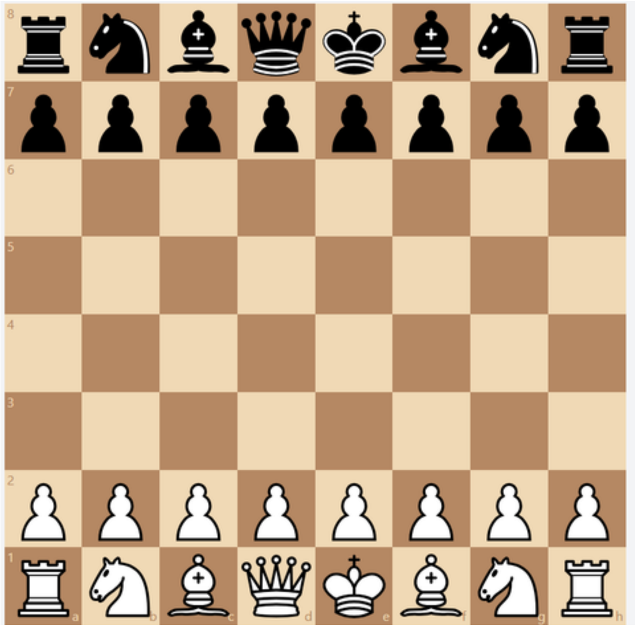
C. Play with Bots:

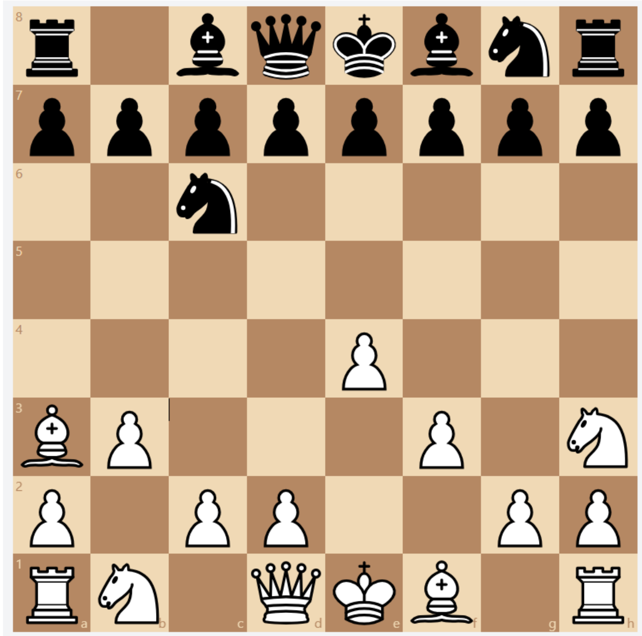
AI has different levels of intelligence.

Suggest moves to the player.



Interface when starting to play





III. How to Play

1. Objective

The objective of chess is to checkmate the opponent's king.

A king is checkmated when it is attacked and has no legal move to escape.

The game can also end in a draw under certain conditions.

2. Setup

Chess is played on an 8x8 board with alternating light and dark squares.

Each player starts with 16 pieces:

1 King ♔

1 Queen ♕

2 Rooks ♖

2 Bishops ♗

2 Knights ♘

8 Pawns ♙

The board must be arranged so that each player has a light square in the lower right corner.

The pieces are placed as follows (from left to right for each player):

Rook – Knight – Bishop – Queen – King – Bishop – Knight – Rook, with the Pawns in the front row.

White moves first.

3. How to play

3.1 How the pieces move

- Each type of piece moves in a specific way:
- Pawn: Moves one square forward, but captures diagonally. Can move two squares on the first move.
- Rook: Moves any number of squares horizontally or vertically.
- Knight: Moves in an L-shape (two squares in one direction, then one square perpendicular). Can jump over pieces.
- Bishop: Moves any number of squares diagonally.
- Queen: Moves any number of squares horizontally, vertically, or diagonally. The most powerful piece.
- King: Moves one square in any direction. The king cannot move into check.

3.2 Special moves

Castling: A move involving the king and a rook, allowing the king to move two squares toward the rook and the rook to jump over the king. This can only be done if:

Neither the king nor the rook have moved.

There are no pieces between them.

The king is not in check and does not move through or into check.

En Passant: A special pawn capture occurs when a pawn moves two squares forward and stops next to an opponent's pawn. The opponent can capture the pawn as if it had only moved one square forward, but this must be done immediately.

Promotion of a pawn: When a pawn reaches the 8th rank, it must be promoted to any other piece (except the king), usually the queen.

3.3 How to win the game

- Checkmate: The game ends when a king is in check and has no legal move to escape.
- Surrender: A player can surrender at any time.
- Time control: In timed games, a player loses if the time runs out.

3.4 Draw Conditions

A game can end in a draw under the following conditions:

- Deadlock: Players have no legal moves, but their king is not in check.
- Triplet: The same position occurs three times with the same possible moves.
- Insufficient pieces: Neither side has enough pieces to checkmate (e.g. king vs king, king and knight vs king).
- 50-move rule: If 50 consecutive moves occur without a move by a pawn or capture, the game is a draw.
- Mutual agreement: Players can agree to a draw.

IV. Development tools.

Visual Studio Code: Visual studio tool is used to code related to front-end of web

game. Including front-end of home page, chess board, login and register.

Visual Studio Code

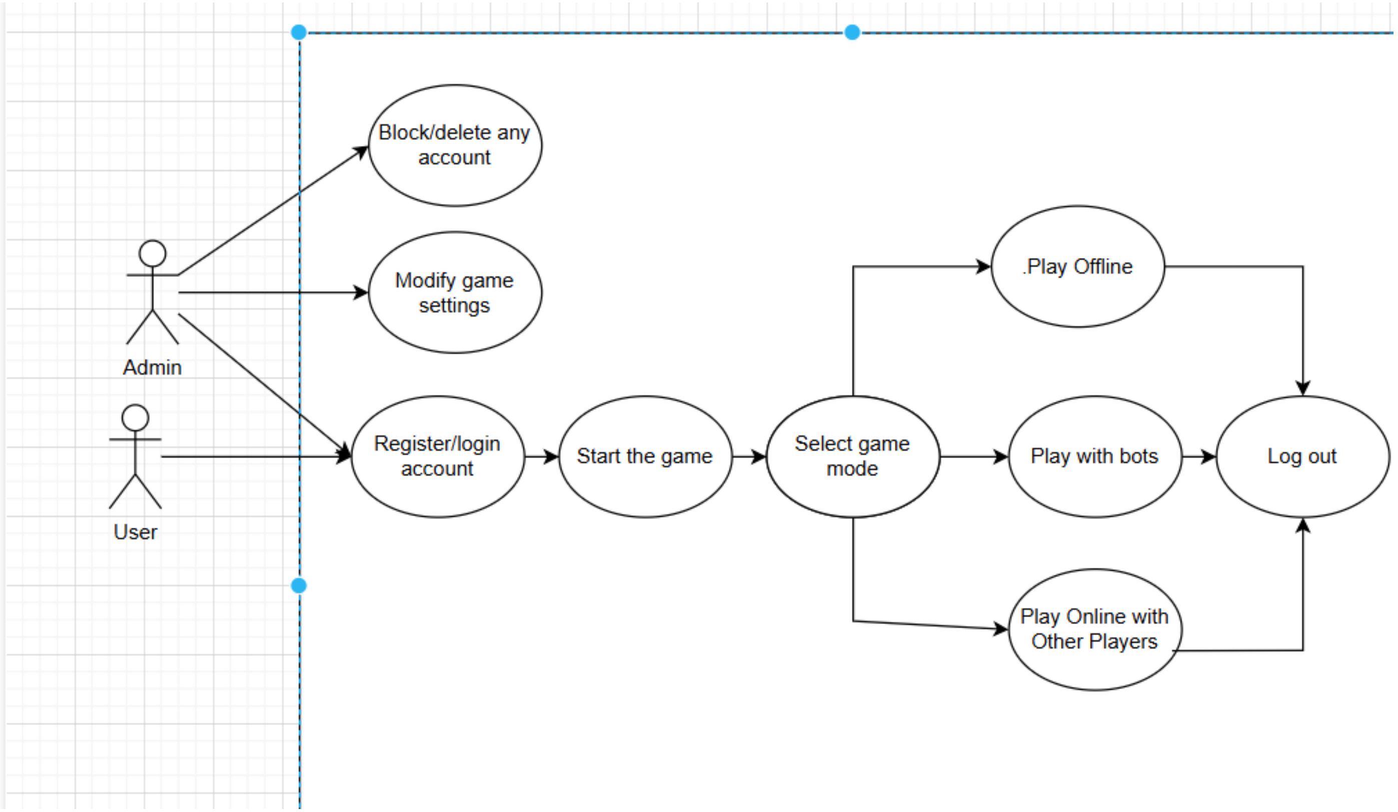
IntelliJ IDEA: IntelliJ tool is used to code the back-end for the game, such as:

being able to play the game, being able to register and log into the game.

MongoDB: MongoDB is used to store game data into the Database, including

storing game registration accounts, game play history and player achievements.

V. Usecase



Block/delete any account
Modify game settings
Admin
User
Register/login account
Start the game
Select game mode
.Play Offline
Play with bots
Play Online with Other Players
Log out

Explanation of the Use Case Diagram of Chess Website

The Use Case diagram above describes the main functions of the online chess system, including the roles of User and Admin.

1. Actors in the system

User: User registers an account, logs in, plays chess, selects the game mode and logs out.

Admin: Has the right to manage user accounts and game settings.

2. Main Use Cases

A. Functions for Users

Register/Login account

Users need to register or log in to access the system.

Start the game

After logging in, users can start playing chess.

Select game mode

Users can choose one of the following game modes:

- Play Offline: Play alone or compete with AI.
- Play with bots: Play with a computer with integrated AI.
- Play Online with Other Players: Play online with real players.

Log out

After completing the game or not wanting to continue using it, the player can log out of the system.

B. Functions for Admin

Block/Delete any account

Admin has the right to manage player accounts, can block or delete accounts that violate the rules.

Modify game settings

Admin can change game settings, including game rules, AI level, time limit, etc.

3. Description of the flow

- Player registers or logs in to the account.
- After logging in, the player starts the game.
- Player selects the game mode (play with bots, play offline, play with others).
- When the game ends, the player can continue playing or log out.
- Admin has the right to manage player accounts and edit game settings.

VI.Commercial and Development Direction.

The commercial viability of a Chess game site depends on its broad appeal to casual players and educational institutions, creating opportunities for monetization through a freemium model, in-app purchases, and advertising. To maximize revenue, the site can incorporate features such as premium themes, advanced AI opponents, and social sharing options. Moving forward, the focus should be on enhancing the user experience through cross-platform compatibility, sophisticated AI, and multiplayer capabilities. Regular updates, such as seasonal themes and community engagement initiatives, can boost player retention. Additionally, incorporating user feedback and accessibility features will ensure that the game remains inclusive and appealing to a wide audience, ultimately driving long-term growth and success.